Table 1. Metadata for acoustical monitoring sites at Gates of the Arctic National Park and Preserve (GAAR). Dates reflect each site's recorded NVSPL deployment period(s).

Site	Site Name	Dates	Vegetation	Wilderness	Elevation (m)	Latitude	Longitude
GAARAMLA	NA		NA	NA	NA	NA	NA
GAARANAK	NA		NA	NA	NA	NA	NA
GAARARRI	NA		NA	NA	NA	NA	NA
GAARFLOR	NA		NA	NA	NA	NA	NA
GAARHAMM	NA		NA	NA	NA	NA	NA
GAARITKI	NA		NA	NA	NA	NA	NA
GAARKURU	NA		NA	NA	NA	NA	NA
GAARMWAL	NA		NA	NA	NA	NA	NA
GAARNARV	NA		NA	NA	NA	NA	NA
GAARNNUT	NA		NA	NA	NA	NA	NA
GAARNUTU	NA		NA	NA	NA	NA	NA
GAARNWAL	NA		NA	NA	NA	NA	NA
GAARPALA	NA		NA	NA	NA	NA	NA
GAARSELB	NA		NA	NA	NA	NA	NA
GAARUPNO	NA		NA	NA	NA	NA	NA
GAARWALK	NA		NA	NA	NA	NA	NA

Table 2. Summary of sound levels at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2016. For each site, includes the 90th percentile (LA10), existing (LA50), natural (LAnat), and 10th percentile (LA90) ambient sound levels (dB re 20 uPa, A−weighted broadband, 12.5−20,000 Hz) during the day (0700−1900) and night (1900−0700).

Time	SiteID	LA10	LA50	LAnat	LA90
Day	GAARANAK	47.7	46.4	46.3	45.0
Night	GAARANAK	48.1	47.0	47.0	46.1

Table 3. Number of days of audio analyzed for the Listening Center and SPLAT detection methods at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2016 − the basis for the analysis−period text reported in the Methods section.

Type	DaysText
ListeningCenter	NA
SPLAT	9

Table 4. Executive summary for Gates of the Arctic National Park and Preserve (GAAR), Summer 2016: mean overall noise audibility, top contributing noise sources, and ambient sound levels (LA50/LAnat, day and night) by site.

SiteID	Noise	Aircraft	Vehicles	NA 4	LA50 Day	LAnat Day	LA50 Night	LAnat Night
GAARANAK	2	2	0	NA	46.4	46.3	47	47

Table 5. Noise impact (dB, existing minus natural ambient) and Listening Area Reduction (%) by site at Gates of the Arctic National Park and Preserve (GAAR), Summer 2016, for day, night, and the combined 24−hour period.

SiteID	DayImpact	DayLAR	NightImpact	NightLAR	AllImpact	AllLAR
GAARANAK	0.1	2.3	0	0	0.1	1.1

Table 6. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2016.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARANAK	1.1	1	Aircraft	1.1	Jet	2.0	0.2	2.2	148.5	190.0	152.6
GAARANAK	1.2	1	Aircraft	1.2	Prop	11.3	1.4	12.8	104.9	128.5	107.6
GAARANAK	2.0	2	Vehicles	2.0	Vehicle	0.1	0.2	0.3	38.0	38.5	38.3

Table 7. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2016.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	785	890	959	3600	216	3600	3600	3600	509	50	195	113	140	16	101	58	390	237	667	3600	167	52	355	3600	GAARANAK
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	873	782	727	1420	1540	964	1194	1320	1748	3600	3600	3600	3196	3261	3600	GAARANAK
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2748	3600	2932	3600	3164	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARANAK
Mean	2922	2976	3003	3600	1940	3600	3600	3600	2677	820	770	711	795	853	744	801	1114	1388	2535	3600	2088	1177	2018	3600	GAARANAK

Table 8. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2016.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARANAK	Jet	0.3	0.0	0.0	0	0.0	0	0	0	0.0	0.0	0.3	1.0	3.2	1.1	1.0	0.6	0.1	1.0	0.0	0	0.0	0.0	0.9	0
GAARANAK	Prop	0.0	0.1	0.1	0	0.4	0	0	0	0.5	4.2	5.2	4.6	2.1	1.8	3.8	3.6	4.3	1.2	1.8	0	0.2	1.1	3.3	0
GAARANAK	Vehicle	0.0	0.0	0.0	0	0.2	0	0	0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0.0	0

Table 9. Percent time above sound level thresholds for acoustic records collected at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2016. Values show the percent of time ANS−weighted sound levels (20−1,250 Hz) and the full measured frequency range (12.5−20,000 Hz) exceeded 35, 45, 52, and 60 dB (LAeq,1s), by site and day/night.

Site ID	Frequency (Hz)	35 dB Day	45 dB Day	52 dB Day	60 dB Day	35 dB Night	45 dB Night	52 dB Night	60 dB Night
GAARANAK	Full (12.5−20,000)	100	86.8	0	0	100	97.5	0	0
GAARANAK	ANS (20−1,250)	100	59.1	0	0	100	80.1	0	0

Figure 1. GAARANAK: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2016.

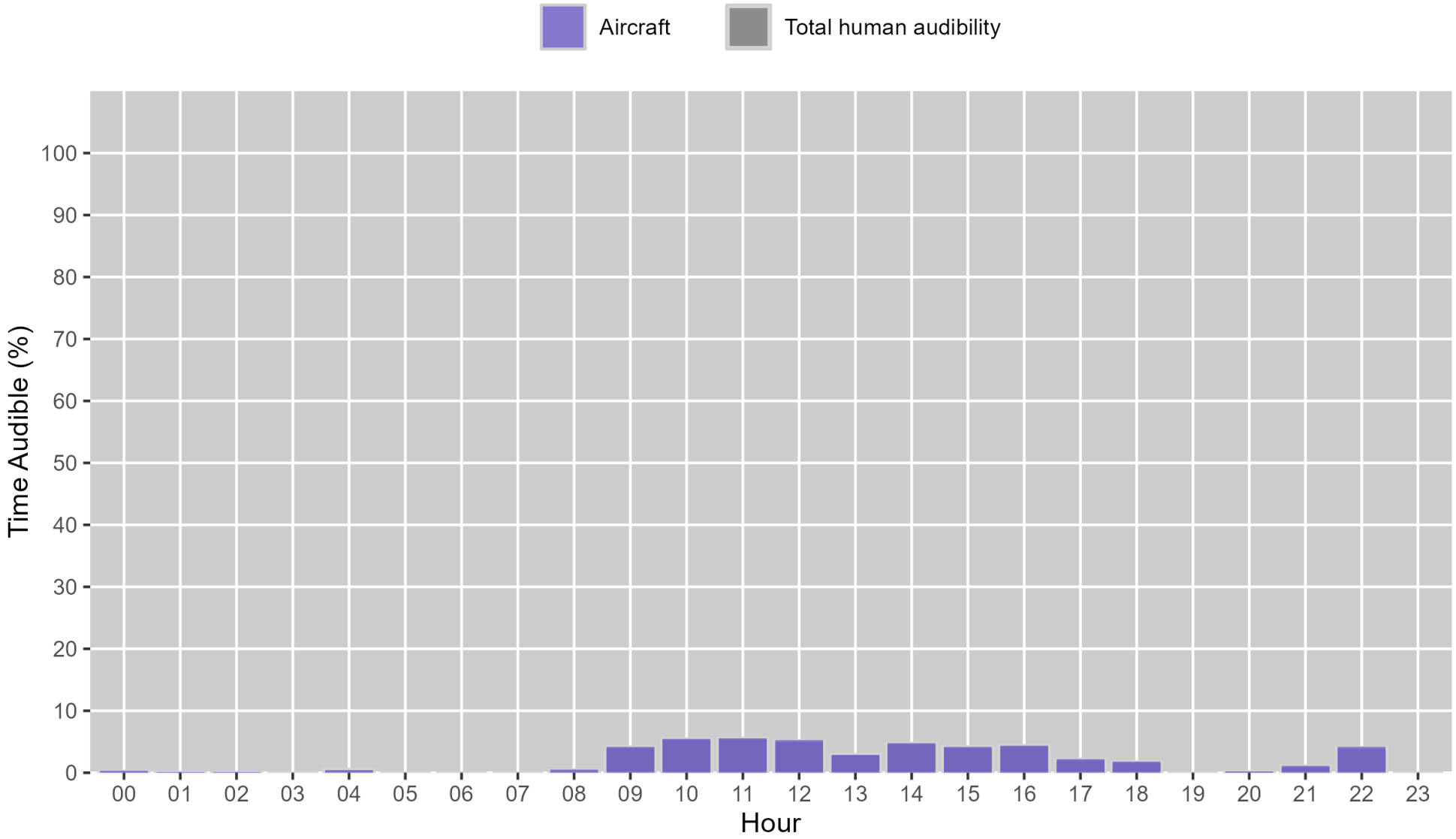


Figure 2. GAARANAK: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2016.

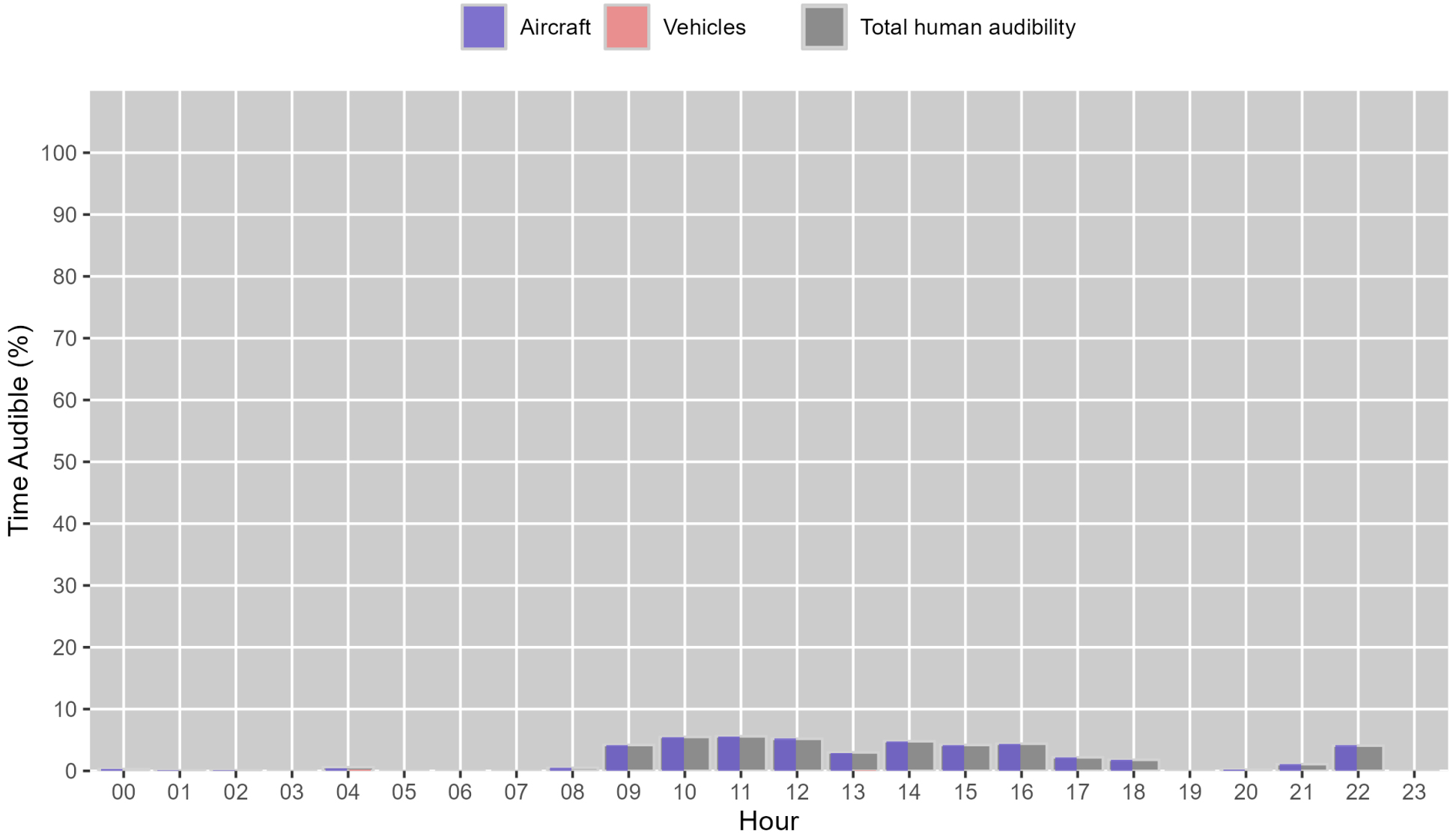


Figure 3. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARANAK in Summer, 2016, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

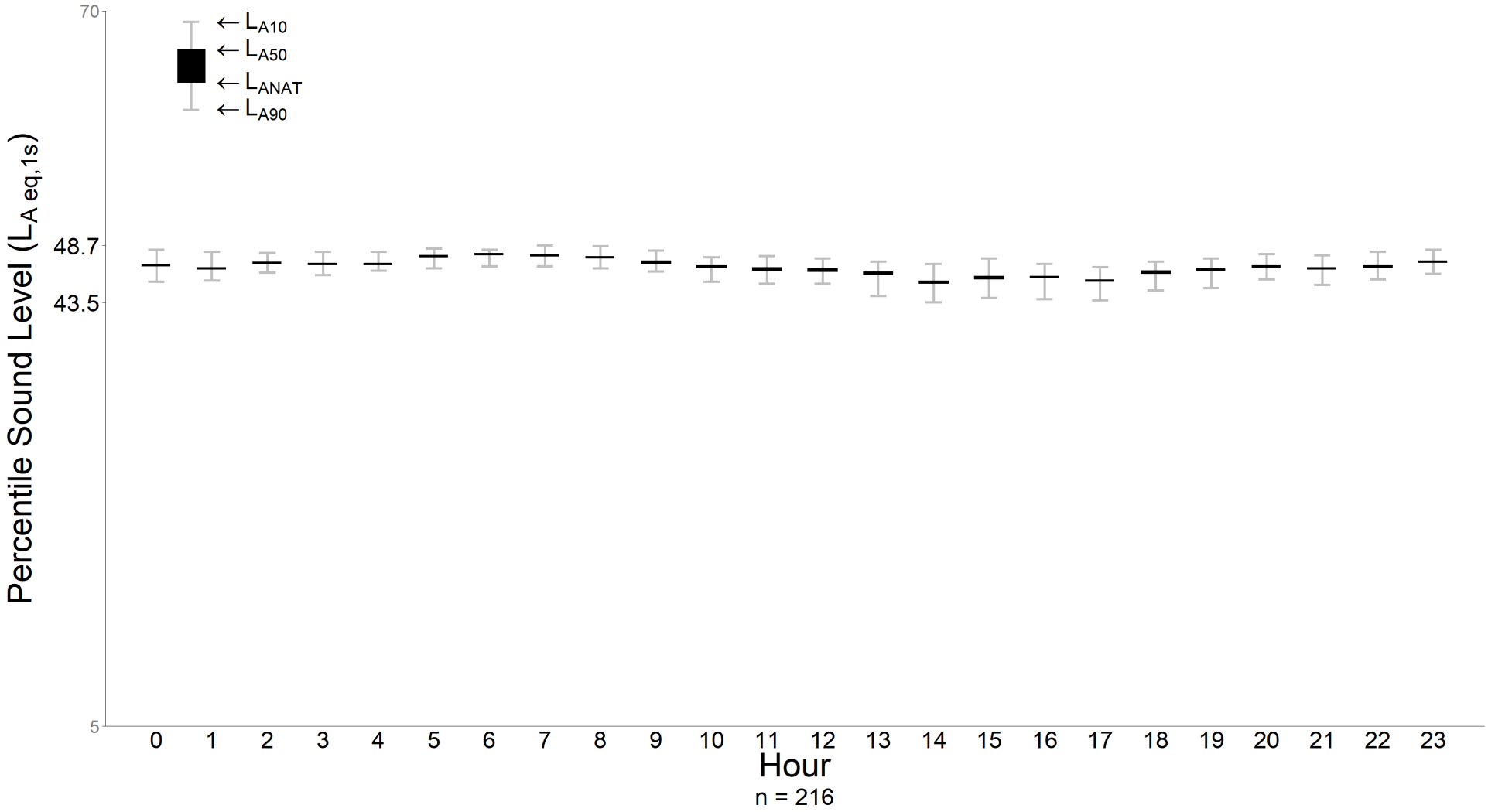


Figure 4. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARANAK in Summer, 2016.

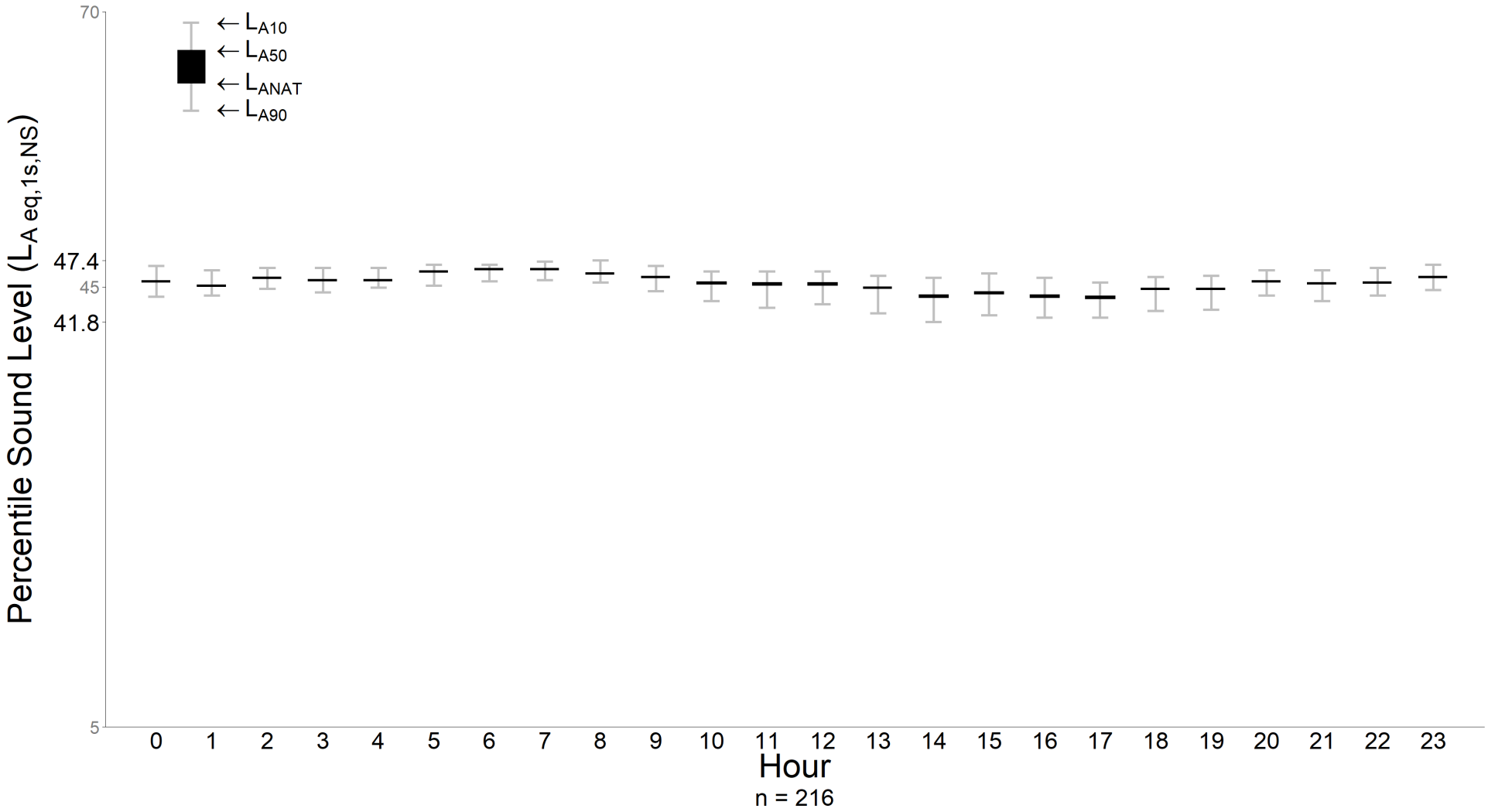


Figure 5. GAARANAK: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2016.

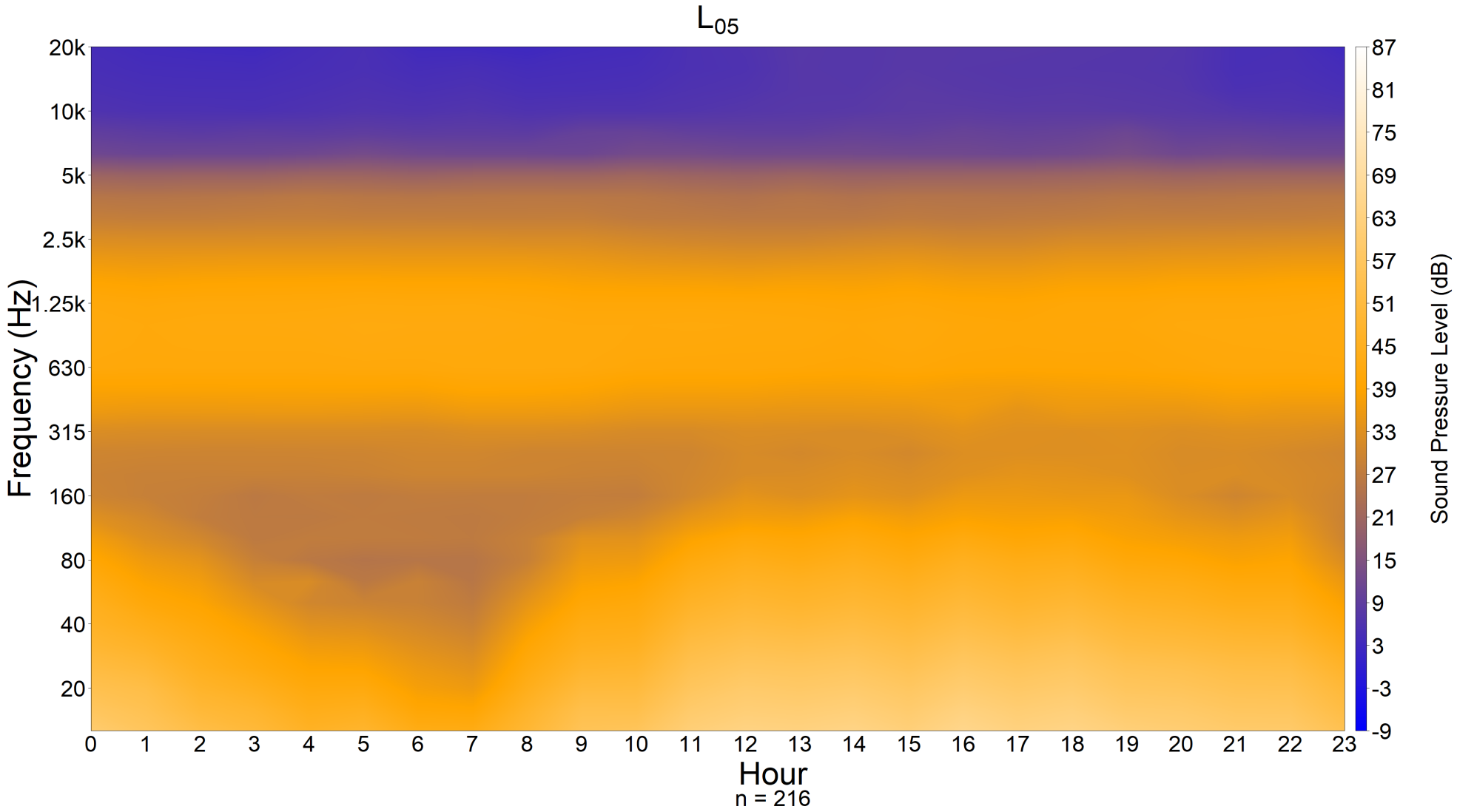


Figure 6. GAARANAK: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2016.

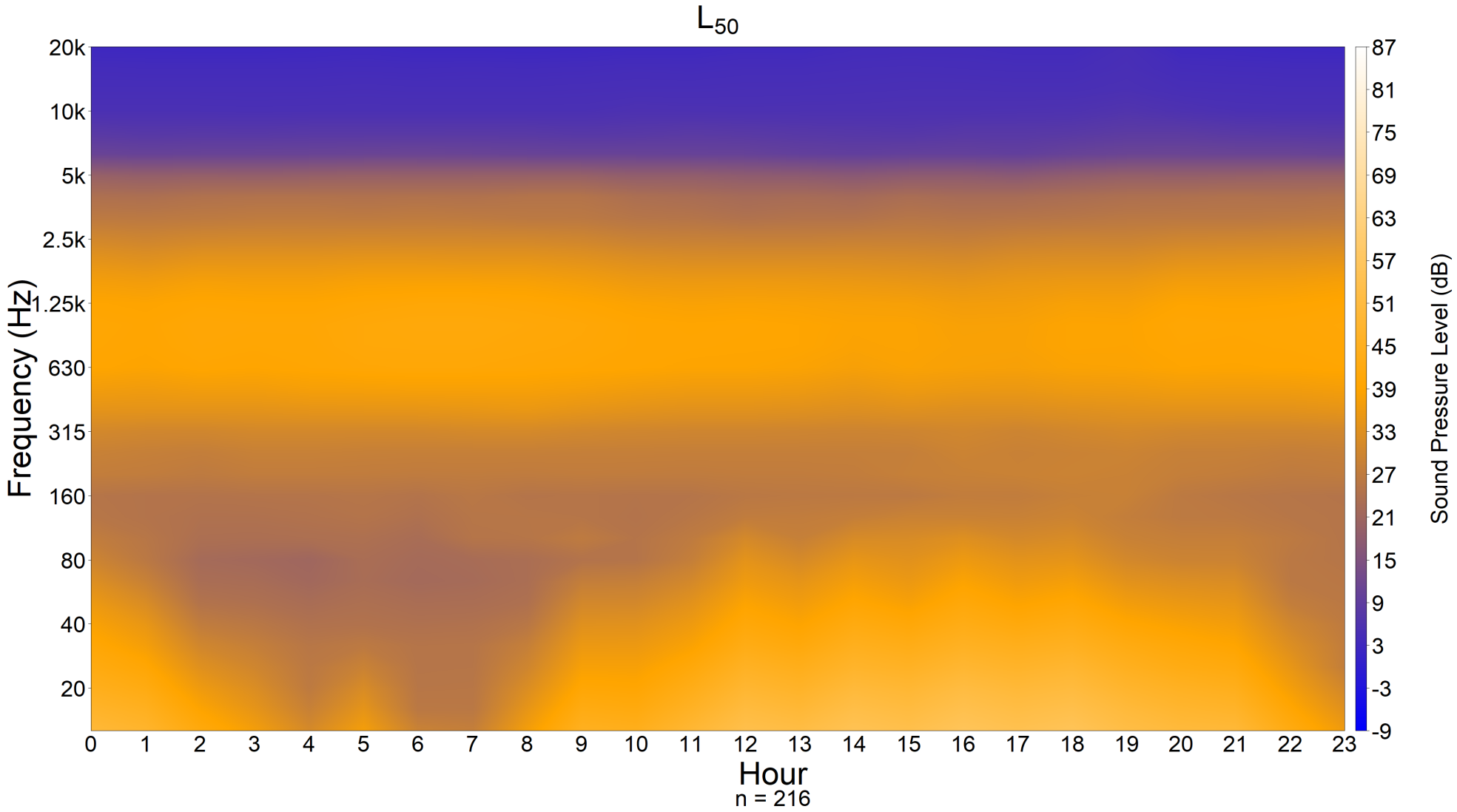


Figure 7. GAARANAK: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2016.

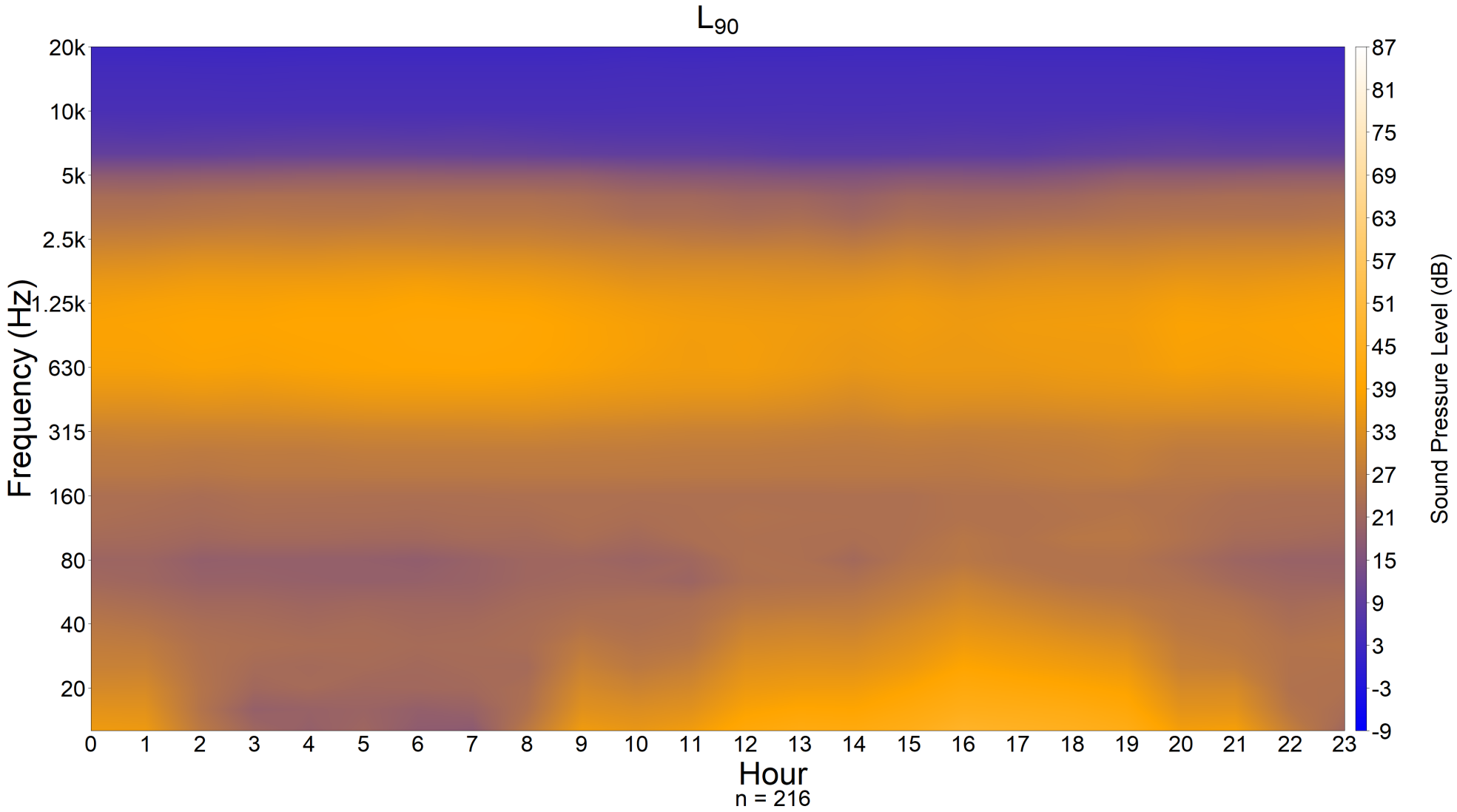


Figure 8. GAARANAK: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2016.

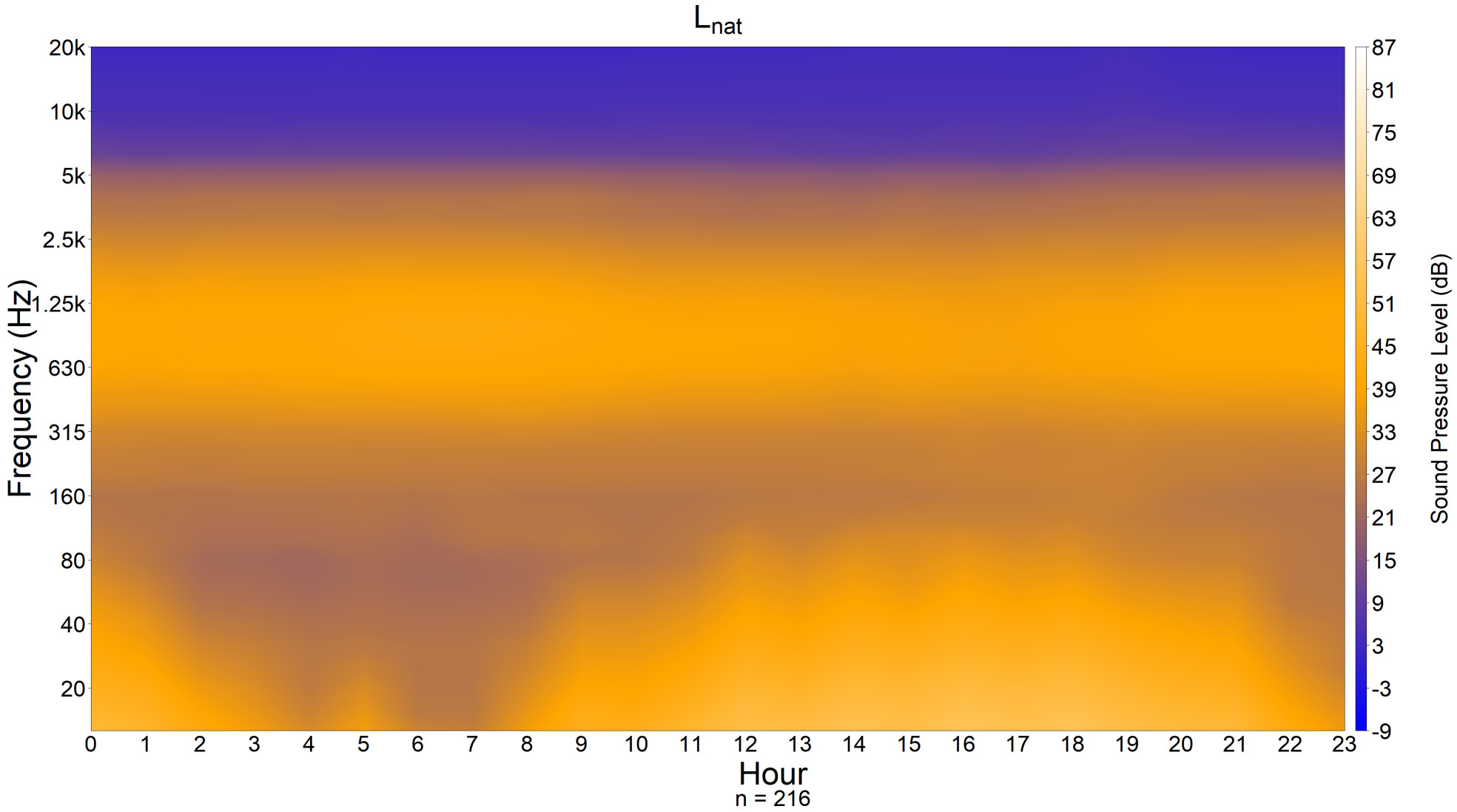


Figure 9. GAARANAK: median and mean duration of noise−free intervals by hour of day, in Summer 2016.

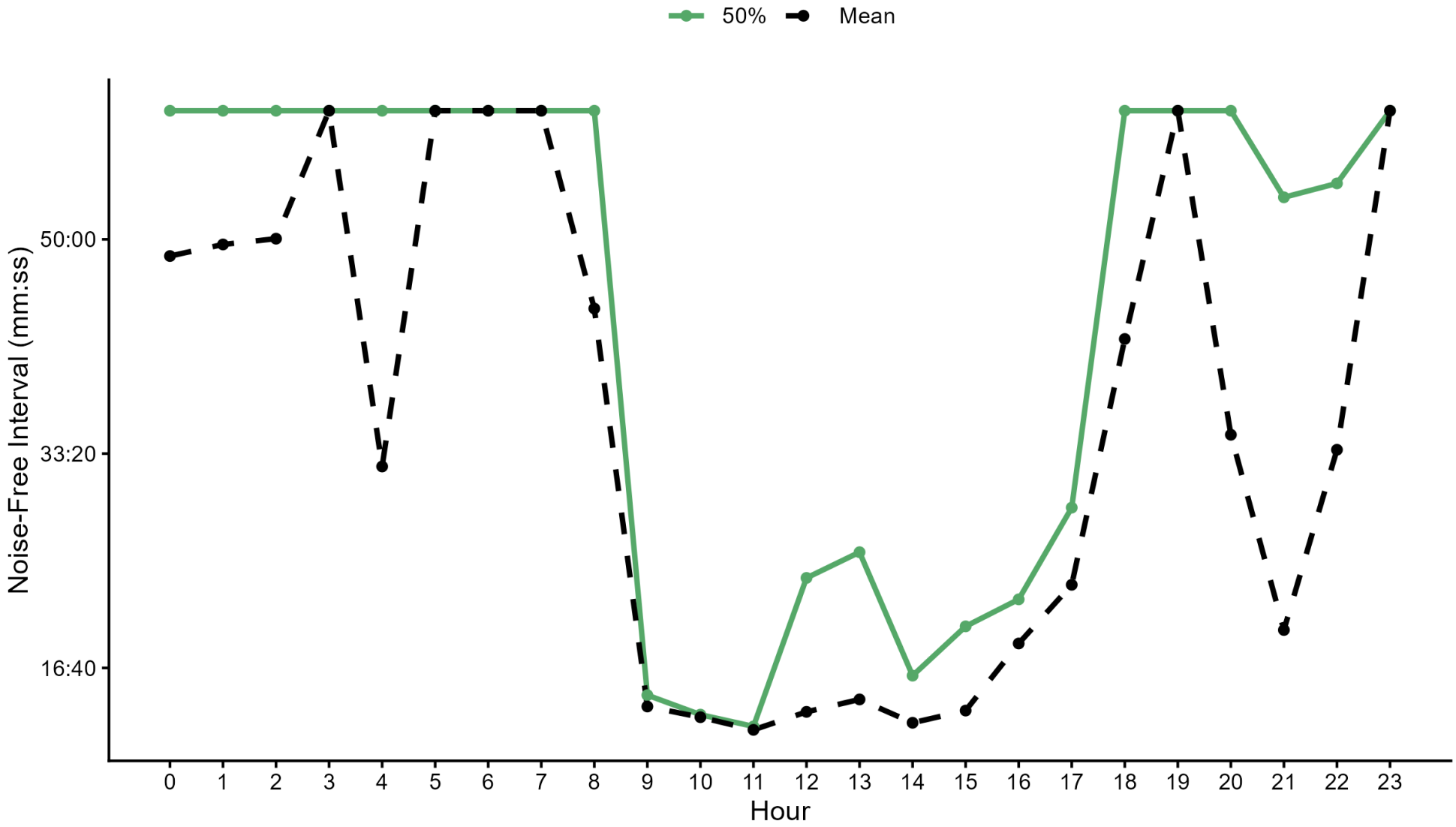


Figure 10. GAARANAK: percent of time audible by sound source category (noise/natural), in Summer 2016.

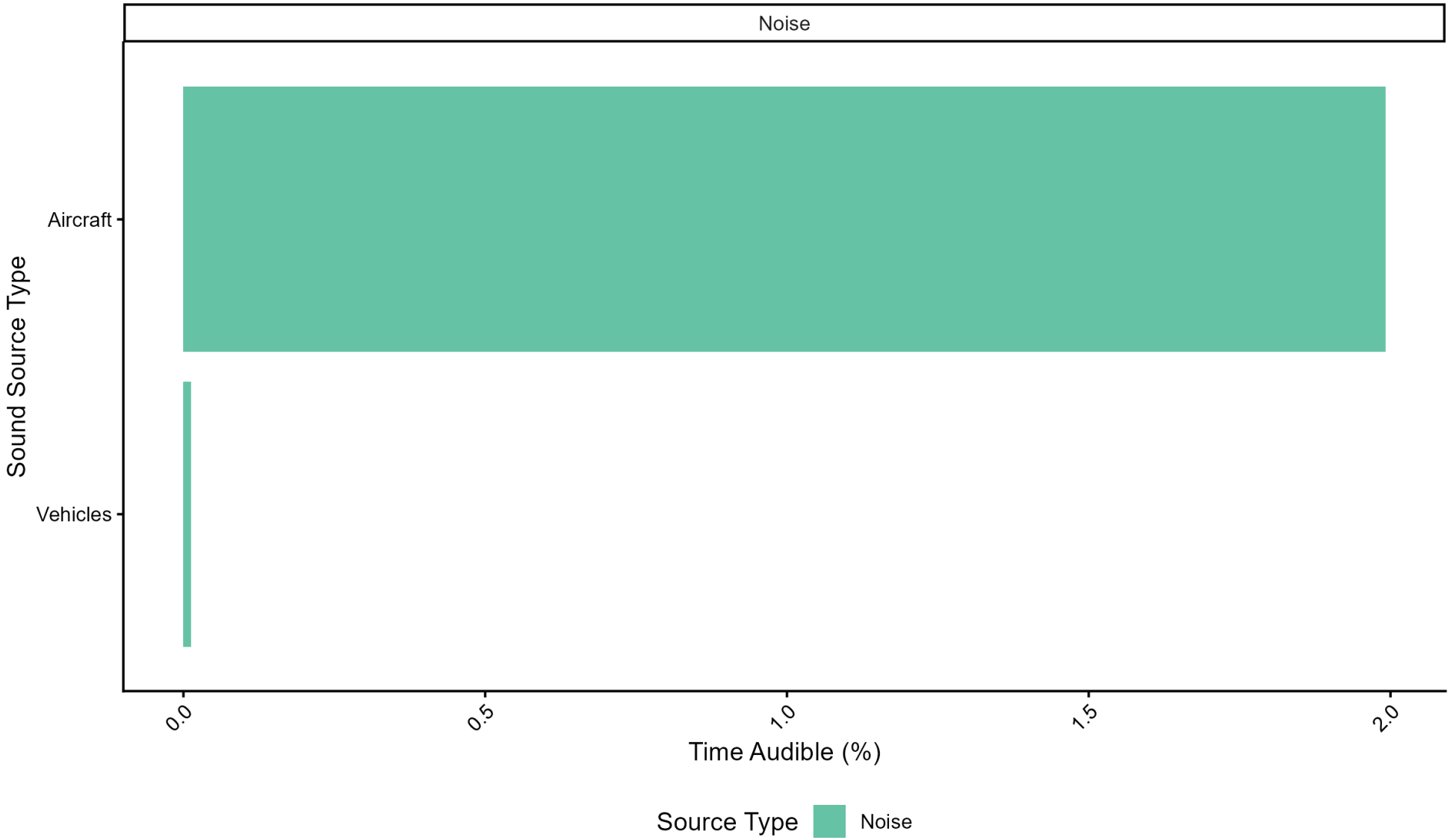


Figure 11. GAARANAK: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2016.

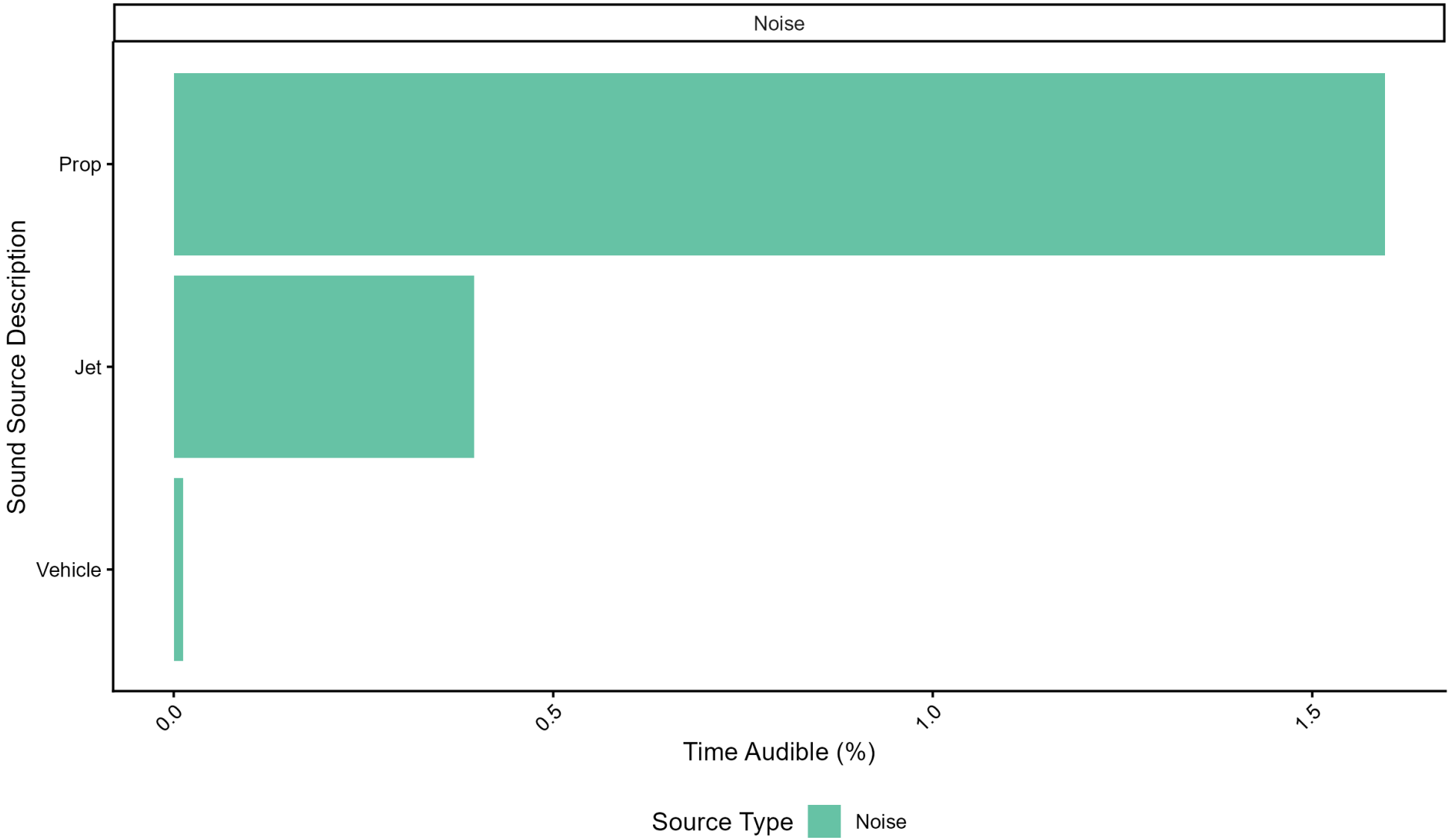


Figure 12. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARANAK in Summer, 2016. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

